OwnDeck - One-Page App Summary

What It Is

OwnDeck is a consumer lifecycle manager for uploading invoices, extracting purchase data, and tracking warranty timelines. It also helps users generate complaint content and recovery details when products fail or go missing.

Who It's For

Primary user: individual consumers who want to organize purchases, warranty records, and support claims. Detailed persona research artifacts: Not found in repo.

What It Does

- JWT-based signup/login and protected app routes for user-specific data.
- Invoice upload from JPG/PNG/WEBP/PDF (up to 10MB), with optional camera capture in the UI.
- OCR extraction from PDFs (pdf-parse) and images (tesseract.js), plus manual fallback fields.
- AI-assisted invoice structuring (OpenAI with fallback parser) and warranty metadata storage in MongoDB.
- Dashboard with search/filter and warranty-state views: Active, Expiring, and Expired.
- AI complaint toolkit: email/tweet/call script generation, tone control, and complaint-refinement chat.
- Lost-item mode with share card (masked contact), and a daily pre-expiry alert cron (SMTP or console fallback).

How It Works (Architecture)

- Frontend (React + Vite + Tailwind): pages for home/auth/dashboard/upload/product detail; API client attaches Bearer token from localStorage.
- Backend (Express): /api/auth, /api/products, and /api/ai routes; JWT middleware protects user endpoints.
- Upload flow: Multer temp file -> OCR service -> invoice parser service -> storage adapter (Cloudinary or local /uploads) -> Product document in MongoDB.
- AI endpoints produce complaint drafts/chat guidance, lost-card data, and product insights; fallback logic is used when OpenAI is unavailable.
- Scheduler/data operations: node-cron runs daily at 09:00 server time, checks warranty windows, and triggers email alerts via Nodemailer when configured.

How To Run

- Install Node.js 20+, npm 10+, and MongoDB (local or Atlas).
- From project root run: npm install, then npm run install:all.
- Copy env templates: backend/.env.example -> backend/.env and frontend/.env.example -> frontend/.env.
- Set minimum env values: MONGO_URI, JWT_SECRET, FRONTEND_URL, and VITE_API_BASE_URL (OpenAI/Cloudinary/SMTP optional).
- Start both services with npm run dev, then open http://localhost:5173 (backend health: /api/health).